

PR_03.1_PIG

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Apartado B

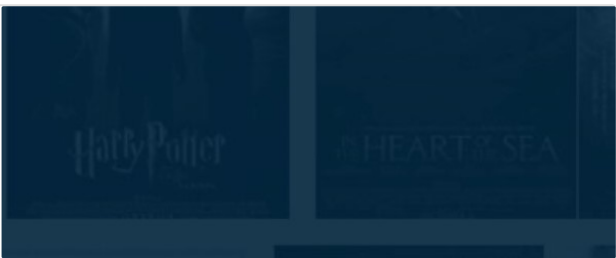
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Trabajaremos durante la práctica con el archivo *u.user* que se encuentra en el siguiente dataset.

MovieLens 100K Dataset

Stable benchmark dataset. 100,000 ratings from 1000 users on 1700 movies

 <https://www.kaggle.com/datasets/prajitdatta/movielens-100k-dataset>



El archivo tiene esta pinta:

```
C: > Users > manel > Downloads > archive > ml-100k > u.user
1 1|24|M|technician|85711
2 2|53|F|other|94043
3 3|23|M|writer|32067
4 4|24|M|technician|43537
5 5|33|F|other|15213
6 6|42|M|executive|98101
7 7|57|M|administrator|91344
8 8|36|M|administrator|05201
9 9|29|M|student|01002
```

user id | age | gender | occupation | zip code

Apartado A

Paso 0. Iniciar la consola de PIG y carga de datos

```
maria_dev@sandbox-hdp:~$ pig
25/11/18 16:47:57 INFO pig.ExecTypeProvider: Trying ExecType : LOCAL
25/11/18 16:47:57 INFO pig.ExecTypeProvider: Trying ExecType : MAPREDUCE
25/11/18 16:47:57 INFO pig.ExecTypeProvider: Trying ExecType : TEZ_LOCAL
25/11/18 16:47:57 INFO pig.ExecTypeProvider: Trying ExecType : TEZ
25/11/18 16:47:57 INFO pig.ExecTypeProvider: Picked TEZ as the ExecType
2025-11-18 16:47:58,032 [main] INFO org.apache.pig.Main - Apache Pig version 0.16.0.2.6.5.0-292 (rUnve
rsioned directory) compiled May 11 2018, 07:56:28
2025-11-18 16:47:58,033 [main] INFO org.apache.pig.Main - Logging error messages to: /home/maria_dev/p
ig_1763484478030.log
2025-11-18 16:47:58,063 [main] INFO org.apache.pig.impl.util.Utils - Default bootup file /home/maria_d
ev/.pigbootup not found
2025-11-18 16:47:58,492 [main] INFO org.apache.pig.backend.hadoop.executionengine.HExecutionEngine - C
onnecting to hadoop file system at: hdfs://sandbox-hdp.hortonworks.com:8020
2025-11-18 16:47:59,003 [main] INFO org.apache.pig.PigServer - Pig Script ID for the session: PIG-defa
ult-49387586-aa5a-4f1f-a398-b4d97c529d8c
2025-11-18 16:47:59,261 [main] INFO org.apache.hadoop.yarn.client.api.impl.TimelineClientImpl - Timeli
ne service address: http://sandbox-hdp.hortonworks.com:8188/ws/v1/timeline/
2025-11-18 16:47:59,331 [main] INFO org.apache.pig.backend.hadoop.PigATSCClient - Created ATS Hook
grunt>
```

Iniciamos PIG

```
grunt> usuarios = LOAD '/user/maria_dev/u.user' USING PigStorage('|') AS (id:int, age:int, gender:chara
rray, occupation:chararray, zip:chararray);
```

Primero cargamos la información en una variable

1. Total de Hombres y Mujeres

Vamos a contar cuántos usuarios hay por género.

```
grupos_genero = GROUP usuarios BY gender;
conteo_genero = FOREACH grupos_genero GENERATE group AS genero, COUNT(usuarios) AS total;
DUMP conteo_genero;
```



```

maria_dev@sandbox-hdp:~
grunt> grupos_genero = GROUP usuarios BY gender;
grunt> conteo_genero = FOREACH grupos_genero GENERATE group AS genero, COUNT(usuarios) AS total;
grunt> DUMP conteo_genero
2025-11-18 17:03:56,603 [main] INFO org.apache.pig.tools.pigstats.ScriptState - Pig features used in the script: GROUP_BY
2025-11-18 17:03:56,633 [main] INFO org.apache.pig.data.SchemaTupleBackend - Key [pig.schematuple] was not set... will not generate code.
2025-11-18 17:03:56,661 [main] INFO org.apache.pig.newplan.logical.optimizer.LogicalPlanOptimizer - {RULES_ENABLED=[AddForEach, ColumnMapKeyPrune, ConstantCalculator, GroupByConstParallelSetter, LimitOptimizer, LoadTypeCastInserter, MergeFilter, MergeForEach, PartitionFilterOptimizer, PredicatePushdownOptimizer, PushDownForEachFlatten, PushUpFilter, SplitFilter, StreamTypeCastInserter]}
2025-11-18 17:03:56,693 [main] INFO org.apache.pig.impl.util.SpillableMemoryManager - Selected heap (PS Old Gen) of size 699400192 to monitor. collectionUsageThreshold = 489580128, usageThreshold = 489580128
2025-11-18 17:03:56,745 [main] INFO org.apache.pig.backend.hadoop.executionengine.tez.TezLauncher - Tez staging directory is /tmp/maria_dev/staging and resources directory is /tmp/temp-261890944
2025-11-18 17:03:56,769 [main] INFO org.apache.pig.backend.hadoop.executionengine.tez.plan.TezCompiler - File concatenation threshold: 100 optimistic? false
2025-11-18 17:03:56,793 [main] INFO org.apache.pig.backend.hadoop.executionengine.util.CombinerOptimizerUtil - Choosing to move algebraic foreach to combiner
2025-11-18 17:03:56,851 [main] INFO org.apache.pig.builtin.PigStorage - Using PigTextInputFormat
2025-11-18 17:03:56,856 [main] INFO org.apache.hadoop.mapreduce.lib.input.FileInputFormat - Total input paths to process : 1
2025-11-18 17:03:56,856 [main] INFO org.apache.pig.backend.hadoop.executionengine.util.MapRedUtil - Total input paths to process : 1
2025-11-18 17:03:56,903 [main] INFO org.apache.pig.backend.hadoop.executionengine.util.MapRedUtil - Total input paths (combined) to process : 1
2025-11-18 17:03:57,713 [main] INFO org.apache.pig.backend.hadoop.executionengine.tez.TezJobCompiler - Local resource: joda-time-2.9.4.jar
2025-11-18 17:03:57,713 [main] INFO org.apache.pig.backend.hadoop.executionengine.tez.TezJobCompiler - Local resource: antlr-runtime-3.4.jar
2025-11-18 17:03:57,713 [main] INFO org.apache.pig.backend.hadoop.executionengine.tez.TezJobCompiler - Local resource: pig-0.16.0-2.6.5.0-292-core-h2.jar
2025-11-18 17:03:57,713 [main] INFO org.apache.pig.backend.hadoop.executionengine.tez.TezJobCompiler - Local resource: automaton-1.11-8.jar
2025-11-18 17:03:57,828 [main] INFO org.apache.pig.backend.hadoop.executionengine.tez.TezDagBuilder - For vertex - scope-28: parallelism=1, memory=384, java opts=-Xmx384m

```

```

maria_dev@sandbox-hdp:~
TotalLaunchedTasks: 2
  FileBytesRead: 110
  FileBytesWritten: 78
  HdfsBytesRead: 22628
  HdfsBytesWritten: 22
SpillableMemoryManager spill count: 0
  Bags proactively spilled: 0
  Records proactively spilled: 0

DAG Plan:
Tez vertex scope-28 → Tez vertex scope-29,
Tez vertex scope-29

Vertex Stats:
VertexId Parallelism TotalTasks InputRecords ReduceInputRecords OutputRecords FileBytesRead FileBytesWritten HdfsBytesRead HdfsBytesWritten Alias Feature Outputs
scope-28 1 1 943 0 943 32 78 22
628 0 conteo_genero,grupos_genero,usuarios
scope-29 1 1 0 2 2 78 0
0 22 conteo_genero GROUP_BY hdfs://sandbox-hdp.hortonworks.com:8020/tmp/temp420001292/tmp-1434748521,

Input(s):
Successfully read 943 records (22628 bytes) from: "/user/maria_dev/u.user"

Output(s):
Successfully stored 2 records (22 bytes) in: "hdfs://sandbox-hdp.hortonworks.com:8020/tmp/temp420001292/tmp-1434748521"

2025-11-18 17:04:20,109 [main] INFO org.apache.hadoop.mapreduce.lib.input.FileInputFormat - Total input paths to process : 1
2025-11-18 17:04:20,109 [main] INFO org.apache.pig.backend.hadoop.executionengine.util.MapRedUtil - Total input paths to process : 1
(F,273)
(M,678)
grunt>

```

2. Top 10 Ocupaciones más Frecuentes

Buscamos los trabajos más comunes, ordenados de mayor a menor.

```

grupos_ocupacion = GROUP usuarios BY occupation;
conteo_ocupacion = FOREACH grupos_ocupacion GENERATE group AS ocupacion, COUNT(usuarios) AS total;
ocupaciones_ordenadas = ORDER conteo_ocupacion BY total DESC;
top_10_ocupaciones = LIMIT ocupaciones_ordenadas 10;
DUMP top_10_ocupaciones;

```



```

maria_dev@sandbox-hdp:~
grunt> grupos_ocupacion = GROUP usuarios BY occupation;
grunt> conteo_ocupacion = FOREACH grupos_ocupacion GENERATE group AS ocupacion, COUNT(usuarios) AS total;
grunt> ocupaciones_ordenadas = ORDER conteo_ocupacion BY total DESC;
grunt> top_10_ocupaciones = LIMIT ocupaciones_ordenadas 10;
grunt> DUMP top_10_ocupaciones;

```

```

maria_dev@sandbox-hdp:~
ead HdfsBytesWritten Alias      Feature Outputs
scope-100      -1      1      0      10      10      409      257
0      0      ORDER_BY
scope-111      1      1      0      10      10      257      0
0      175      LIMIT      hdfs://sandbox-hdp.hortonworks.com:8020/tmp/temp420001292/tmp884279817,
scope-78      1      1      943      0      943      32      367      22
628      0      conteo_ocupacion,grupos_ocupacion,usuarios
scope-79      1      1      0      21      42      399      651
0      0      conteo_ocupacion,ocupaciones_ordenadas      GROUP_BY,SAMPLER
scope-88      1      1      0      21      1      254      140
0      0
scope-98      1      1      22      0      21      569      409
0      0      ocupaciones_ordenadas

Input(s):
Successfully read 943 records (22628 bytes) from: "/user/maria_dev/u.user"

Output(s):
Successfully stored 10 records (175 bytes) in: "hdfs://sandbox-hdp.hortonworks.com:8020/tmp/temp420001292/tmp884279817"

2025-11-18 17:05:57,457 [main] INFO      org.apache.hadoop.mapreduce.lib.input.FileInputFormat - Total input paths to process : 1
2025-11-18 17:05:57,457 [main] INFO      org.apache.pig.backend.hadoop.executionengine.util.MapRedUtil - Total input paths to process : 1
(student,196)
(other,105)
(educator,95)
(administrator,79)
(engineer,67)
(programmer,66)
(librarian,51)
(writer,45)
(executive,32)
(scientist,31)
grunt>

```

3. Edad Media por Género

Agrupamos por género y calculamos la media.

```

edad_genero = FOREACH grupos_genero GENERATE group AS genero, AVG(usuarios.age) AS media_edad;
DUMP edad_genero;

```



```

    maria_dev@sandbox-hdp:~
    grunt> edad_media_genero = FOREACH grupos_genero GENERATE group AS genero, AVG(usuarios.age) AS media_edad;
    grunt> DUMP edad_media_genero;

```

```

    maria_dev@sandbox-hdp:~
    HdfsBytesWritten: 34
    SpillableMemoryManager spill count: 0
    Bags proactively spilled: 0
    Records proactively spilled: 0

    DAG Plan:
    Tez vertex scope-162  →    Tez vertex scope-163,
    Tez vertex scope-163

    Vertex Stats:
    VertexId Parallelism TotalTasks  InputRecords  ReduceInputRecords  OutputRecords  FileBytesRead  FileBytesWritten  HdfsBytesR
    ead HdfsBytesWritten Alias      Feature Outputs
    scope-162      1      1      943
    628      0 edad_media_genero,grupos_genero,usuarios
    scope-163      1      1      0
    0      34 edad_media_genero  GROUP_BY      hdfs://sandbox-hdp.hortonworks.com:8020/tmp/temp420001292/tmp196728234
    7,

    Input(s):
    Successfully read 943 records (22628 bytes) from: "/user/ maria_dev/u.user"

    Output(s):
    Successfully stored 2 records (34 bytes) in: "hdfs://sandbox-hdp.hortonworks.com:8020/tmp/temp420001292/tmp1967282347"

    2025-11-18 17:07:55,450 [main] INFO  org.apache.hadoop.mapreduce.lib.input.FileInputFormat - Total input paths to process : 1
    2025-11-18 17:07:55,450 [main] INFO  org.apache.pig.backend.hadoop.executionengine.util.MapRedUtil - Total input paths to proc
    ess : 1
    (F,33.81318681318681)
    (M,34.149253731343286)
    grunt>

```

4. Edad media por Ocupaci3n

Promedio de edad para cada profesi3n.

```

edad_ocupacion = FOREACH grupos_ocupacion GENERATE group AS ocupacion, AVG(usuarios.age) AS media_edad;
DUMP edad_ocupacion;

```



```

    maria_dev@sandbox-hdp:~
    grunt> edad_media_ocupacion = FOREACH grupos_ocupacion GENERATE group AS ocupacion, AVG(usuarios.age) AS media_edad;
    grunt> DUMP edad_media_ocupacion;

```

```

    maria_dev@sandbox-hdp:~
    0          508 edad_media_ocupacion      GROUP_BY      hdfs://sandbox-hdp.hortonworks.com:8020/tmp/temp420001292/tmp-169268804,
    Input(s):
    Successfully read 943 records (22628 bytes) from: "/user/maria_dev/u.user"
    Output(s):
    Successfully stored 21 records (508 bytes) in: "hdfs://sandbox-hdp.hortonworks.com:8020/tmp/temp420001292/tmp-169268804"
    2025-11-18 17:08:36,209 [main] INFO  org.apache.hadoop.mapreduce.lib.input.FileInputFormat - Total input paths to process : 1
    2025-11-18 17:08:36,209 [main] INFO  org.apache.pig.backend.hadoop.executionengine.util.MapRedUtil - Total input paths to process : 1
    (none,26.555555555555557)
    (other,34.523809523809526)
    (artist,31.392857142857142)
    (doctor,43.57142857142857)
    (lawyer,36.75)
    (writer,36.31111111111111)
    (retired,63.07142857142857)
    (student,22.081632653061224)
    (educator,42.01052631578948)
    (engineer,36.38805970149254)
    (salesman,35.666666666666664)
    (executive,38.71875)
    (homemaker,32.57142857142857)
    (librarian,40.0)
    (marketing,37.61538461538461)
    (scientist,35.54838709677419)
    (healthcare,41.5625)
    (programmer,33.121212121212125)
    (technician,33.148148148148145)
    (administrator,38.74683544303797)
    (entertainment,29.22222222222222)
    grunt>

```

5. Creamos el fichero del script (.pig)

Creamos un script y lo cargamos con el siguiente contenido.

El enunciado pide guardar esto e un archivo `.pig` y que el resultado no salga por pantalla (DUMP) sino que se guarde en una carpeta (STORE).

- 1. Creamos un archivo en windows con extensión `.pig`.
- 2. En VSCode creamos el script (explicado en la imagen).


```
Windows PowerShell X + v
PS C:\Users\manel\Documents\CURSO DE ESPECIALIZACIÓN - INTELIGENCIA ARTIFICIAL\BigData> New-Item -PATH 'C:\Users\manel\Documents\CURSO DE ESPECIALIZACIÓN - INTELIGENCIA ARTIFICIAL\BigData\ analisis_usuarios.pig' -ItemType File

Directory: C:\Users\manel\Documents\CURSO DE ESPECIALIZACIÓN - INTELIGENCIA ARTIFICIAL\BigData

Mode                LastWriteTime         Length Name
----                -
-a----           18/11/2025   18:17             0 analisis_usuarios.pig

PS C:\Users\manel\Documents\CURSO DE ESPECIALIZACIÓN - INTELIGENCIA ARTIFICIAL\BigData> |
```

```
File Edit Selection ... Search
Restricted Mode is intended for safe code browsing. Trust this window to enable all features. Manage Learn More

analisis_usuarios.pig
C: > Users > manel > Documents > CURSO DE ESPECIALIZACIÓN - INTELIGENCIA ARTIFICIAL > BigData > Prácticas > PR_03.1_PIG > analisis_usuarios.pig
1 /* Carga de datos */
2 usuarios = LOAD '/user/maria_dev/u.user' USING PigStorage('|')
3   AS (id:int, age:int, gender:chararray, occupation:chararray, zip:chararray);
4
5 /* 1. Total de hombres y mujeres */
6 grupos_genero = GROUP usuarios BY gender;
7 conteo_genero = FOREACH grupos_genero GENERATE group AS genero, COUNT(usuarios) AS total;
8 /* Guardamos en la carpeta 'genero' dentro de 'pig_usuarios' */
9 STORE conteo_genero INTO '/user/maria_dev/pig_usuarios/genero';
10
11 /* 2. Las 10 ocupaciones más frecuentes */
12 grupos_ocupacion = GROUP usuarios BY occupation;
13 conteo_ocupacion = FOREACH grupos_ocupacion GENERATE group AS ocupacion, COUNT(usuarios) AS total;
14 ocupaciones_ordenadas = ORDER conteo_ocupacion BY total DESC;
15 top_10_ocupaciones = LIMIT ocupaciones_ordenadas 10;
16 STORE top_10_ocupaciones INTO '/user/maria_dev/pig_usuarios/ocupaciones';
17
18 /* 3. Edad media por generos */
19 edad_media_genero = FOREACH grupos_genero GENERATE group AS genero, AVG(usuarios.age) AS edad_promedio;
20 STORE edad_media_genero INTO '/user/maria_dev/pig_usuarios/edad_genero';
21
22 /* 4. Edad media por ocupaciones */
23 edad_media_ocupacion = FOREACH grupos_ocupacion GENERATE group AS ocupacion, AVG(usuarios.age) AS edad_promedio;
24 STORE edad_media_ocupacion INTO '/user/maria_dev/pig_usuarios/edad_ocupacion';
25
```

Una vez subido a ambari el archivo, lo ejecutamos con pig.

```
maria_dev@sandbox-hdp:~ X + v
[maria_dev@sandbox-hdp ~]$ hdfs dfs -get /user/maria_dev/ analisis_usuarios.pig
get: ' analisis_usuarios.pig': File exists
[maria_dev@sandbox-hdp ~]$ pig analisis_usuarios.pig
```


Estas imágenes muestran el contenido de la carpeta creada.

```
maria_dev@sandbox-hdp:~$ hdfs dfs -ls /user/maria_dev/pig_usuarios
Found 4 items
drwxr-xr-x - maria_dev hdfs      0 2025-11-18 17:23 /user/maria_dev/pig_usuarios/edad_genero
drwxr-xr-x - maria_dev hdfs      0 2025-11-18 17:23 /user/maria_dev/pig_usuarios/edad_ocupacion
drwxr-xr-x - maria_dev hdfs      0 2025-11-18 17:23 /user/maria_dev/pig_usuarios/genero
drwxr-xr-x - maria_dev hdfs      0 2025-11-18 17:23 /user/maria_dev/pig_usuarios/ocupaciones
maria_dev@sandbox-hdp ~]$

maria_dev@sandbox-hdp ~]$ hdfs dfs -cat /user/maria_dev/pig_usuarios/edad_genero/part-v001-o000-r-00000
F      33.81318681318681
M      34.149253731343286
maria_dev@sandbox-hdp ~]$
```

Apartado B

Creamos la variable ventas en pig a partir de nuestro nuevo CSV (subido a ambari).

```
maria_dev@sandbox-hdp:~$ grunt> ventas = LOAD '/user/maria_dev/retail_sales_dataset.csv' USING PigStorage(',')
>> AS (trans_id:int, date:chararray, cust_id:chararray, gender:chararray,
>>    age:int, category:chararray, quantity:int, price:int, total:int);
```

1. Filtrado por Rango de edad

Filtrar y ver resultados


```

  maria_dev@sandbox-hdp:~
  X + v
  - □ X

  grunt> clientes_mayores30 = FILTER ventas BY age > 30;
  grunt> muestras_filtradas = LIMIT clientes_mayores30 10;
  grunt> DUMP muestras_filtradas;
```

```

  maria_dev@sandbox-hdp:~
  X + v
  - □ X

  FileBytesWritten: 388
  HdfsBytesRead: 51673
  HdfsBytesWritten: 563
  SpillableMemoryManager spill count: 0
  Bags proactively spilled: 0
  Records proactively spilled: 0

  DAG Plan:
  Tez vertex scope-35      →      Tez vertex scope-37,
  Tez vertex scope-37

  Vertex Stats:
  VertexId Parallelism TotalTasks  InputRecords  ReduceInputRecords  OutputRecords  FileBytesRead  FileBytesWritten  HdfsBytesRead  HdfsBytesWri
  tten Alias      Feature  Outputs
  scope-35      1      1      16      0      10      0      388      51673
  0 clientes_mayores30,muestras_filtradas,ventas
  scope-37      1      1      10      0      10      388      0      0
  563 muestras_filtradas LIMIT hdfs://sandbox-hdp.hortonworks.com:8020/tmp/temp1020225705/tmp325999190,

  Input(s):
  Successfully read 16 records (51673 bytes) from: "/user/maria_dev/retail_sales_dataset.csv"

  Output(s):
  Successfully stored 10 records (563 bytes) in: "hdfs://sandbox-hdp.hortonworks.com:8020/tmp/temp1020225705/tmp325999190"

  2025-11-18 17:36:33,029 [main] INFO org.apache.hadoop.mapreduce.lib.input.FileInputFormat - Total input paths to process : 1
  2025-11-18 17:36:33,030 [main] INFO org.apache.pig.backend.hadoop.executionengine.util.MapRedUtil - Total input paths to process : 1
  (1,2023-11-24,CUST001,Male,34,Beauty,3,50,150)
  (3,2023-01-13,CUST003,Male,50,Electronics,1,30,30)
  (4,2023-05-21,CUST004,Male,37,Clothing,1,500,500)
  (6,2023-04-25,CUST006,Female,45,Beauty,1,30,30)
  (7,2023-03-13,CUST007,Male,46,Clothing,2,25,50)
  (9,2023-12-13,CUST009,Male,63,Electronics,2,300,600)
  (10,2023-10-07,CUST010,Female,52,Clothing,4,50,200)
  (12,2023-10-30,CUST012,Male,35,Beauty,3,25,75)
  (14,2023-01-17,CUST014,Male,64,Clothing,4,30,120)
  (15,2023-01-16,CUST015,Female,42,Electronics,4,500,2000)
  grunt>
```

Calcular porcentaje

```

  maria_dev@sandbox-hdp:~
  X + v
  - □ X

  grunt> conteo_mayores = GROUP clientes_mayores30 ALL;
  grunt> total_mayores = FOREACH conteo_mayores GENERATE COUNT(clientes_mayores30) AS total_mayores;
  grunt> DUMP total_mayores;
```



```
Success!

DAG 0:
      Name: PigLatin:DefaultJobName-0_scope-2
      ApplicationId: job_1763485262225_0003
      TotalLaunchedTasks: 2
      FileBytesRead: 99
      FileBytesWritten: 67
      HdfsBytesRead: 51673
      HdfsBytesWritten: 7
      SpillableMemoryManager spill count: 0
      Bags proactively spilled: 0
      Records proactively spilled: 0

DAG Plan:
Tez vertex scope-82  →  Tez vertex scope-83,
Tez vertex scope-83

Vertex Stats:
VertexId Parallelism TotalTasks  InputRecords  ReduceInputRecords  OutputRecords  FileBytesRead  FileBytesWritten  HdfsBytesRead  HdfsBytesWri
tten Alias      Feature  Outputs
scope-82      1          1          1801          0          727          32          67          51673
0 clientes_mayores30, conteo_mayores, total_mayores, ventas
scope-83      1          1           0          1           1          67           0           0
7 total_mayores      GROUP_BY      hdfs://sandbox-hdp.hortonworks.com:8020/tmp/temp1020225705/tmp-1366175368,

Input(s):
Successfully read 1801 records (51673 bytes) from: "/user/maria_dev/retail_sales_dataset.csv"

Output(s):
Successfully stored 1 records (7 bytes) in: "hdfs://sandbox-hdp.hortonworks.com:8020/tmp/temp1020225705/tmp-1366175368"

2025-11-18 17:38:14,976 [main] INFO  org.apache.hadoop.mapreduce.lib.input.FileInputFormat - Total input paths to process : 1
2025-11-18 17:38:14,976 [main] INFO  org.apache.pig.backend.hadoop.executionengine.util.MapRedUtil - Total input paths to process : 1
(727)
grunt> █
```

Qué porcentaje del total de transacciones corresponde a mayores de 30?

El conteo total son 1000 y de mayores de 30 son 727, es el 27.3%.

Transformación de Campos

```
grunt> datos_transformados = FOREACH ventas GENERATE
grunt>   trans_id,
>>   UPPER(gender) AS gender_upper, /* Género en mayúsculas */
>>   category,
>>   (price * quantity * 0.90) AS importe_descuento; /* 10% de descuento ficticio */
2025-11-18 17:55:24,500 [main] WARN  org.apache.pig.newplan.BaseOperatorPlan - Encountered Warning IMPLICIT_CAST_TO_DOUBLE 1 time(s).
grunt>
grunt> muestras_transformadas = LIMIT datos_transformados 20;
2025-11-18 17:55:24,620 [main] WARN  org.apache.pig.newplan.BaseOperatorPlan - Encountered Warning IMPLICIT_CAST_TO_DOUBLE 1 time(s).
grunt> DUMP muestras_transformadas;█
```



```

2025-11-18 17:55:27,471 [main] INFO org.apache.pig.newplan.logical.optimizer.LogicalPlanOptimizer - {RULES_ENABLED=[AddForEach, ColumnMapKeyPrune, ConstantCalculator, GroupByConstParallelSetter, LimitOptimizer, LoadTypeCastInserter, MergeFilter, MergeForEach, PartitionFilterOptimizer, PredicatePushdownOptimizer, PushDownForEachFlatten, PushUpFilter, SplitFilter, StreamTypeCastInserter]}
2025-11-18 17:55:27,475 [main] INFO org.apache.pig.newplan.logical.rules.ColumnPruneVisitor - Columns pruned for ventas: $1, $2, $4, $8
2025-11-18 17:55:27,491 [main] INFO org.apache.hadoop.mapreduce.lib.output.FileOutputCommitter - File Output Committer Algorithm version is 1
2025-11-18 17:55:27,491 [main] INFO org.apache.hadoop.mapreduce.lib.output.FileOutputCommitter - FileOutputCommitter skip cleanup _temporary folders under output directory:false, ignore cleanup failures: false
2025-11-18 17:55:27,505 [main] INFO org.apache.pig.data.SchemaTupleBackend - Key [pig.schematuple] was not set... will not generate code.
2025-11-18 17:55:27,512 [main] WARN org.apache.pig.data.SchemaTupleBackend - SchemaTupleBackend has already been initialized
2025-11-18 17:55:27,512 [main] INFO org.apache.pig.builtin.PigStorage - Using PigTextInputFormat
2025-11-18 17:55:27,513 [main] INFO org.apache.hadoop.mapreduce.lib.input.FileInputFormat - Total input paths to process : 1
2025-11-18 17:55:27,513 [main] INFO org.apache.pig.backend.hadoop.executionengine.util.MapRedUtil - Total input paths to process : 1
2025-11-18 17:55:27,568 [main] INFO org.apache.hadoop.mapreduce.lib.output.FileOutputCommitter - Saved output of task 'attempt__0001_m_000001_1' to hdfs://sandbox-hdp.hortonworks.com:8020/tmp/temp-285598847/tmp883696906/_temporary/0/task_0001_m_000001
2025-11-18 17:55:27,982 [main] WARN org.apache.pig.data.SchemaTupleBackend - SchemaTupleBackend has already been initialized
2025-11-18 17:55:27,985 [main] INFO org.apache.hadoop.mapreduce.lib.input.FileInputFormat - Total input paths to process : 1
2025-11-18 17:55:27,985 [main] INFO org.apache.pig.backend.hadoop.executionengine.util.MapRedUtil - Total input paths to process : 1
(, GENDER, Product Category,)
(1, MALE, Beauty, 135.0)
(2, FEMALE, Clothing, 900.0)
(3, MALE, Electronics, 27.0)
(4, MALE, Clothing, 450.0)
(5, MALE, Beauty, 90.0)
(6, FEMALE, Beauty, 27.0)
(7, MALE, Clothing, 45.0)
(8, MALE, Electronics, 90.0)
(9, MALE, Electronics, 540.0)
(10, FEMALE, Clothing, 180.0)
(11, MALE, Clothing, 90.0)
(12, MALE, Beauty, 67.5)
(13, MALE, Electronics, 1350.0)
(14, MALE, Clothing, 180.0)
(15, FEMALE, Electronics, 1800.0)
(16, MALE, Clothing, 1350.0)
(17, FEMALE, Clothing, 90.0)
(18, FEMALE, Electronics, 45.0)
(19, FEMALE, Clothing, 45.0)
grunt>
```

Agrupación y Agregación por Categoría

```

grunt> ventas_por_categoria = GROUP ventas BY category;
2025-11-18 17:56:05,975 [main] WARN org.apache.pig.newplan.BaseOperatorPlan - Encountered Warning IMPLICIT_CAST_TO_DOUBLE 1 time(s).
grunt>
grunt> analisis_categoria = FOREACH ventas_por_categoria GENERATE
grunt>   group AS categoria,
>>   COUNT(ventas) AS num_transacciones,
>>   SUM(ventas.total) AS suma_ventas,
>>   AVG(ventas.age) AS edad_media_cliente;
2025-11-18 17:56:06,046 [main] WARN org.apache.pig.newplan.BaseOperatorPlan - Encountered Warning IMPLICIT_CAST_TO_DOUBLE 1 time(s).
grunt>
grunt> orden_ventas = ORDER analisis_categoria BY suma_ventas DESC;
2025-11-18 17:56:06,096 [main] WARN org.apache.pig.newplan.BaseOperatorPlan - Encountered Warning IMPLICIT_CAST_TO_DOUBLE 1 time(s).
grunt> DUMP orden_ventas;
```

```

SpillableMemoryManager spill count: 0
Bags proactively spilled: 0
Records proactively spilled: 0

DAG Plan:
Tez vertex scope-416 → Tez vertex scope-417,
Tez vertex scope-417 → Tez vertex scope-426, Tez vertex scope-436,
Tez vertex scope-426 → Tez vertex scope-436,
Tez vertex scope-436 → Tez vertex scope-438,
Tez vertex scope-438

Vertex Stats:
VertexId Parallelism TotalTasks InputRecords ReduceInputRecords OutputRecords FileBytesRead FileBytesWritten HdfsBytesRead HdfsBytesWri
tten Alias Feature Outputs
scope-416 1 1 1001 0 1001 32 176 51673
0 analisis_categoria,ventas,ventas_por_categoria
scope-417 1 1 0 4 8 208 299 0
0 analisis_categoria,orden_ventas GROUP_BY,SAMPLER
scope-426 1 1 0 4 1 115 139 0
0
scope-436 1 1 5 0 4 355 186 0
0 orden_ventas
scope-438 -1 1 0 4 4 186 0 0
123 ORDER_BY hdfs://sandbox-hdp.hortonworks.com:8020/tmp/temp-285598847/tmp-1083777413,

Input(s):
Successfully read 1001 records (51673 bytes) from: "/user/maria_dev/retail_sales_dataset.csv"

Output(s):
Successfully stored 4 records (123 bytes) in: "hdfs://sandbox-hdp.hortonworks.com:8020/tmp/temp-285598847/tmp-1083777413"

2025-11-18 17:56:41,275 [main] INFO org.apache.hadoop.mapreduce.lib.input.FileInputFormat - Total input paths to process : 1
2025-11-18 17:56:41,275 [main] INFO org.apache.pig.backend.hadoop.executionengine.util.MapRedUtil - Total input paths to process : 1
(Electronics, 342, 156905, 41.73684210526316)
(Clothing, 351, 155580, 41.94871794871795)
(Beauty, 307, 143515, 40.37133550408599)
(Product Category, 0,,)
grunt>
```

Extracción de Categorías Distintas


```

maria_dev@sandbox-hdp:~
grunt> categorias_distintas = FOREACH ventas GENERATE category;
2025-11-18 17:57:24,232 [main] WARN org.apache.pig.newplan.BaseOperatorPlan - Encountered Warning IMPLICIT_CAST_TO_DOUBLE 1 time(s).
grunt> categorias_unicas = DISTINCT categorias_distintas;
2025-11-18 17:57:24,277 [main] WARN org.apache.pig.newplan.BaseOperatorPlan - Encountered Warning IMPLICIT_CAST_TO_DOUBLE 1 time(s).
grunt> DUMP categorias_unicas;
```

!! IMPLICIT_CAST_TO_DOUBLE.

El mensaje de WARN es una “bandera amarilla”que indica que el código podría ser más eficiente o robusto si los tipos de datos se definieran de forma más estricta, es seguro ignorarla.

```

maria_dev@sandbox-hdp:~
DAG 0:
      Name: PigLatin:DefaultJobName-0_scope-14
      ApplicationId: job_1763485262225_0007
      TotalLaunchedTasks: 2
      FileBytesRead: 161
      FileBytesWritten: 129
      HdfsBytesRead: 51673
      HdfsBytesWritten: 69
      SpillableMemoryManager spill count: 0
      Bags proactively spilled: 0
      Records proactively spilled: 0

DAG Plan:
Tez vertex scope-480 → Tez vertex scope-483,
Tez vertex scope-483

Vertex Stats:
VertexId Parallelism TotalTasks InputRecords ReduceInputRecords OutputRecords FileBytesRead FileBytesWritten HdfsBytesRead HdfsBytesWri
tten Alias Feature Outputs
scope-480 1 1 1881 0 1881 32 129 51673
0 categorias_distintas,categorias_unicas,ventas
scope-483 1 1 0 4 4 129 0 0
69 hdfs://sandbox-hdp.hortonworks.com:8020/tmp/temp-285598847/tmp-961712125,

Input(s):
Successfully read 1881 records (51673 bytes) from: "/user/maria_dev/retail_sales_dataset.csv"

Output(s):
Successfully stored 4 records (69 bytes) in: "hdfs://sandbox-hdp.hortonworks.com:8020/tmp/temp-285598847/tmp-961712125"

2025-11-18 17:58:41,883 [main] INFO org.apache.hadoop.mapreduce.lib.input.FileInputFormat - Total input paths to process : 1
2025-11-18 17:58:41,884 [main] INFO org.apache.pig.backend.hadoop.executionengine.util.MapRedUtil - Total input paths to process : 1
(Beauty)
(Clothing)
(Electronics)
(Product Category)
grunt>
```